Chandrashekhar Prajapati

Software Engineer · Frontend (Angular) · Three.js/WebGL CAD · Document tooling

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Summary

Software Engineer with **3+ years** building enterprise Angular applications (TypeScript, RxJS, Signals, PrimeNG). **Primary owner of a custom Three.js/WebGL CAD viewer and editor built from zero** that became the foundation of a core product workflow. Also builds PDF and document tooling, dynamic forms, enterprise authentication and real-time UIs. Now growing into Python backend (FastAPI, Django) and AI engineering (RAG, LLM agents), building an AI-native portfolio in public.

Experience

Software Engineer, VassarDigital.ai · Hyderabad — Jan 2024 – Present

- **Custom CAD viewer/editor (Angular, Three.js, WebGL, Canvas):** evaluated 3 approaches and shipped a DXF rendering engine for 8 entity types plus CAD tables, R-tree spatial indexing, a full editing suite (selection, properties, layers, move/copy/delete, measurement, undo/redo) and DXF/PDF export. It became the foundation of a core product workflow.
- **Led UI technical direction** for the first ~2 months of a railway signalling-plan workflow: scheme drawings digitised into structured JSON, domain assets edited on the CAD canvas, red/green change detection between versions.
- **PDF and document tooling:** progressive PDF viewer streaming pages with PDF.js and HTTP Range Requests (virtualized rendering); document review viewer with precise text/image highlighting, bounding-box overlays and scroll synchronization.
- **Knowledge-base ingestion:** URL, file and bulk upload, permission-based access (RBAC), PDF/HTML rendering and accessibility-tree tooling; HTML bounding-box visualizer built with the backend team.
- **Enterprise authentication:** integrated Keycloak with role-based access control and Azure AD sign-in with MSAL.
- **Dynamic forms:** JSON-driven Reactive Forms with nested FormArrays and custom validators, reused across modules; reusable component architecture and PrimeNG data tables.
- **Performance and real-time:** lazy loading, batching and progressive processing for large DXF, PDF and JSON workloads; live progress for long-running jobs with Server-Sent Events.
- Contributed across 4 product teams; code reviews, debugging and architecture discussions.

Software Engineer, Vassar Labs · Hyderabad — Aug 2023 – Dec 2023

- Built Angular and TypeScript features: reusable UI components, REST API integrations, UI enhancements and production fixes.

Software Engineer Intern, Vassar Labs · Hyderabad — Jan 2023 – Jul 2023

- Developed Angular UI features and REST integrations within the product team.

Project

AI-native portfolio · chandrashekhar.pages.dev — 2026

- Angular 22 (SSR, signals, zoneless) and FastAPI with PostgreSQL + pgvector and Redis; a knowledge graph of skills, projects and roles whose nodes light up when the AI agent cites them.
- Lighthouse 100 in all four categories, 160+ Playwright end-to-end tests, screenshot tests and CI on every change.

Skills

Frontend	Angular (standalone components, Signals), TypeScript, JavaScript (ES6+), RxJS, Reactive and dynamic forms, PrimeNG, HTML5, SCSS/CSS
CAD & graphics	Three.js, WebGL, Canvas/SVG, DXF, R-tree spatial indexing, undo/redo, DXF/PDF export
Documents	PDF.js, HTTP Range Requests, virtualized rendering, text and bounding-box highlighting
APIs & auth	REST, Server-Sent Events, Keycloak (RBAC), Azure AD / MSAL
Backend & AI	Python, FastAPI, Django, PostgreSQL, LLMs, RAG, LangChain, LangGraph, MCP (growing)
Fundamentals	Data structures & algorithms, C++, Git, Linux, Playwright

Education

Master of Computer Applications (MCA), Computer Software Engineering · National Institute of Technology (NIT) Jamshedpur — 2020 – 2023

Certifications

HackerRank: Problem Solving (Intermediate), Problem Solving (Basic), SQL (Basic) · Workshops: Full Stack Web Development (5 days), Python Programming (6 days)